

LINEAR GRAPHS: PLOTTING $Y = MX + C$

Name: _____

Class/Sec: _____

Date: _____

1. LESSON CONCEPT RECAP & REAL-WORLD CONNECTION



Meet Rohan's Auto Ride!

Rohan takes an auto-rickshaw to the cricket ground. The driver charges a fixed base fare of ₹23 just to start the meter, plus ₹10 for every kilometer traveled!

Formula: Total Fare = Base Fare + (Rate × Distance) → $y = 10x + 23$

The Magic Equation: $y = mx + c$

- x = Input value (e.g., Distance in km)
- y = Output answer (e.g., Total Fare in ₹)
- m = Steepness / Gradient (rate of change / step size)
- c = Starting Value / y -intercept (where line touches y -axis at $x = 0$)

Coordinate Map Basics:

- **x -axis:** Horizontal line (left to right)
- **y -axis:** Vertical line (bottom to top)
- **Origin $(0,0)$:** Where the two axes cross
- **Point Address:** Written as (x, y)

2. GUIDED CALCULATION EXAMPLE: $Y = 2X + 1$

Find y by substituting values of x into the equation to get coordinate pairs (x, y) :

Input (x)	Calculation ($2x + 1$)	Output (y)	Coordinate Point (x, y)
0	$2(0) + 1 = 0 + 1$	1	$(0, 1)$
1	$2(1) + 1 = 2 + 1$	3	$(1, 3)$
2	$2(2) + 1 = 4 + 1$	5	$(2, 5)$
3	$2(3) + 1 = 6 + 1$	7	$(3, 7)$

3. PRACTICE QUESTIONS

Q1. Warm-Up Challenge: Predict the Y-Value

If $y = 3x - 2$, what is the value of y when $x = 2$?

[] A) 6

[] B) 4

[] C) 8

[] D) 10

Q2. Identify Gradient (m) and Y-Intercept (c)

For each equation below, identify the gradient (m) and the y -intercept (c):

a) $y = 4x + 7$ → Gradient (m) = _____, Y-Intercept (c) = _____

b) $y = 6x - 3$ → Gradient (m) = _____, Y-Intercept (c) = _____

Q3. Complete the Table & Find Coordinates

Complete the calculation table below for the line $y = 2x + 3$:

Input (x)	Calculation ($2x + 3$)	Output (y)	Point (x, y)
0	$2(0) + 3$		$(0, \quad)$
1	$2(1) + 3$		$(1, \quad)$
2	$2(2) + 3$		$(2, \quad)$
3	$2(3) + 3$		$(3, \quad)$

Q4. Real-World Application: Rohan's Return Trip

On his return trip home, Rohan takes a different taxi that charges a base fare of ₹15 and a rate of ₹8 per kilometer ($y = 8x + 15$).

a) What is the total fare (y) if Rohan travels $x = 5$ kilometers?

b) Write down the coordinate point (x, y) for this 5 km trip: _____

Q5. Horizontal and Vertical Special Lines

Fill in the blanks for these special line equations:

a) For a horizontal line passing through $y = 4$, the equation is $y =$ _____

b) For a vertical line passing through $x = 3$, the equation is $x =$ _____

Q6. Plotting Practice on Coordinate Grid

Using the table from Question 3 for $y = 2x + 3$, list all four coordinate points you found, and state which quadrant they fall in (assume all positive values are in Quadrant I):

Q7. True or False Statement Check

- a) The point $(0, 0)$ is called the origin. []
- b) In $y = mx + c$, c represents how fast y grows for every 1 step increase in x . []
- c) For the line $y = 5x + 2$, the line crosses the y -axis at $y = 2$. []
- d) A line with equation $y = 4x$ has a y -intercept of $c = 0$. []

Q8. Challenge Question: Reverse Engineering the Equation

A line passes through $(0, 4)$ and has a gradient/slope of $m = 3$. Write down the full equation of this line in $y = mx + c$ form:

Equation: $y =$ _____